Exercise 1.

- 4 pockets - not distinct
- 5 books - not distinct
- 5 books = 5 stars

  4 pockets = 4 - 1 = 3 bars

  $$\binom{5+3}{3} = \binom{8}{3} = \frac{8!}{3!5!} = \frac{40{,}320}{6\times120} = \frac{40{,}320}{720} = 56$$

Exercise 2.

- 10 stars and 3 bars

$$\binom{10+3}{3} = \binom{13}{3} = \frac{13!}{3!10!} = 286$$

286 terms

$$= \binom{10}{a,b,c,d} x^a y^b z^c w^d$$

$$= \binom{10}{0,2,7,1} x^0 y^2 z^7 w^1$$

$$= \frac{10!}{0!2!7!1!} y^2 z^7 w$$

$$= \frac{2{,}628{,}800}{1\times2\times5{,}040\times1} y^2 z^7 w$$

$$= \frac{3{,}628{,}800}{10{,}080} y^2 z^7 w$$

$$= 360 y^2 z^7 w$$

COEFFICIENT: 360

Exercise 3.

- 5 distinct piles = 5 - 1 = 4 bars

(a)

- 20 bricks = 20 stars

$$\binom{20+4}{4} = \binom{24}{4} = \frac{24!}{20!4!} = 10,626$$

(b)

- pile 1 has at least 1 brick

- 19 stars

$\binom{19+4}{4} = \binom{23}{4} = \frac{23!}{4!19!} = 8,855$

(c)

- pile 1 having exactly 1 brick
- 3 bars now

$\binom{19+3}{3} = \binom{22}{3} = \frac{22!}{3!19!} = 1,540$

(d)

- bricks all distinct

$\binom{20}{4,4,4,4,4} = \frac{20!}{(4!)^5} = 305,540,235,000$

Exercise 4.

- 6 chips labeled 1, 2, 3, 4, 5, 6
- 4 people called A, B, C, D

(a) $6^4 = 1,296$

(b) $\binom{6}{4} = \frac{6!}{4!2!} = 15$

(c)

- 6 chips: 6 - 1 = 5 bars
- 4 people = 4 stars

$\binom{5+4}{5} = \binom{9}{5} = \frac{9!}{5!4!} = 126$

Exercise 5.

- marbles: 5 red, 3 blue, and 2 green
- 5 distinct boxes labeled 1-5

red:

- 5 red marbles = 5 stars
- 5 distinct boxes = 5 - 1 = 4 bars

$\binom{5+4}{4} = \binom{9}{4} = \frac{9!}{4!5!} = 126$

blue:

- 3 blue marbles = 3 stars
- 5 distinct boxes = 5 - 1 = 4 bars

$\binom{3+4}{4} = \binom{7}{4} = \frac{7!}{4!3!} = 35$

green:

- 2 green marbles = 2 stars
- 5 distinct boxes = 5 - 1 = 4 bars

$\binom{2+4}{4} = \binom{6}{4} \frac{6!}{4!2!} = 15$

ways marbles can be distributed in boxes $= 126 \times 35 \times 15 = 66,150$

Exercise 6.

(a)

- non-negative integers: 0+
- 125 stars
- 3 - 1 = 2 bars

$\binom{125+2}{2} = \binom{127}{2} = \frac{127!}{2!125!} = \frac{127\times126}{2} = 8,001$

(b)

- 0 not included
- 1 - 125: 125
- 124 stars
- 3 - 1 = 2 bars

$\binom{124+2}{2} = \binom{126}{2} = \frac{126!}{2!124!} = \frac{126\times125}{2} = 7,875$

(c)

- 125 - 101 = 24
- $x_1 + x_2 + x_3 = 24$
- 24 stars
- 2 bars

$\binom{127}{2} - 3\binom{24+2}{2} = \binom{127}{2} - 3\binom{26}{2} = 8,001 - 3\frac{26!}{2!24!} = 8,001 - 3\frac{26\times 25}{2} = 8,001 - 975 =$
$7,026$

Exercise 7.

(a) n = 6

$\binom{6}{2,4} = \frac{6!}{2!4!} = 15$

$\binom{6}{3,3} = \frac{6!}{3!3!} = 20$

Because the jobs are not distinguishable

$\frac{20}{2!} = 10$

Therefore,

$15 + 20 = 25$

(b) n =7

$\binom{7}{2,5} = \frac{7!}{2!5!} = 21$

$\binom{7}{3,4} = \frac{7!}{3!4!} = 35$

$= 21 + 35 = 56$

(c) $\binom{6}{2,4} = \frac{6!}{2!4!} = 15$

$\binom{6}{3,3} = \frac{6!}{3!3!} = 20$

$= 15 + 20 = 35$

Exercise 8.

$|\Omega| = 6^6 = 46,656$

$P(|A|) =$ 1st 4 rolls even

$= (\frac{1}{2})^4 = \frac{1}{16}$

$P(|B|) =$ last 3 rolls divisible by 3

$= 1 \times (\frac{1}{3})^3 = \frac{1}{27}$

$P(|A \cap B|) = (\frac{1}{2})(\frac{1}{6})^3 = \frac{1}{432}$

$P(|A \cup B|) = \frac{1}{16} + \frac{1}{27} - \frac{1}{432} = \frac{7}{72}$

Exercise 9.

- 8 people enter an elevator
- stops at 4 floors
- picks floors uniformly at random
- $|\Omega| = 4^8 = 65,536$

At least one floor is not selected

$= 4 \times 3^8 - \binom{4}{2} 2^8 + \binom{4}{3} 1^8$

$= 26,244 - \frac{4!}{2!2!} 2^8 + \frac{4!}{3!1!} 1^8$

$= 26,244 - 1,536 + 4$

$= 24,712$

$=> \frac{24,712}{65,536} \approx 0.3771$

Every floor is selected

$= 65,536 - 24,712 = 40,824$

$=> \frac{40,824}{65,536} \approx 0.6229$

Exercise 10.

(a)

- total hands: $\binom{52}{5} = \frac{52!}{5!47!} = 2,598,968$
- number of hands void in one specific suit: $\binom{39}{5} = 575,757$
- single suit void: $P(V_1) = \frac{\binom{39}{5}}{\binom{52}{5}} = \frac{575,757}{2,598,968} \approx 0.2216$
- two suits void: $P(V_1 \cap V_2) = \frac{\binom{26}{5}}{\binom{52}{5}} = \frac{65,780}{2,598,968} \approx 0.0253$
- three suits void: $P(V_1 \cap V_2 \cap V_3) = \frac{\binom{13}{5}}{\binom{52}{5}} = \frac{1,287}{2,598,968} \approx 0.0005$
- fours suits void: $P(V_1 \cap V_2 \cap V_3 \cap V_4) = 0$

$P(V_1 \cup V_2 \cup V_3 \cup V_4) \approx \binom{4}{3} 0.2216 - \binom{4}{2} 0.0253 + \binom{4}{1} 0.0005$

$\approx 4 \times 0.2216 - 6 \times 0.0253 + 4 \times 0.0005$

$\approx 0.8864 - 0.1518 + 0.002$

≈ 0.7366

(b)

- total hands: $\binom{52}{13} = \frac{52!}{13!39!} = 635,013,559,600$
- single suit void: $P(V_1) = \frac{\binom{39}{13}}{\binom{52}{13}} = \frac{8,122,425,444}{635,013,559,600} \approx 0.0128$
- two suits void: $P(V_1 \cap V_2) = \frac{\binom{26}{13}}{\binom{52}{13}} = \frac{10,400,600}{635,013,559,600} \approx 0.00002$
- three suits void: $P(V_1 \cap V_2 \cap V_3) = \frac{\binom{13}{13}}{\binom{52}{13}} = \frac{1}{635,013,559,600} = 1.57476952 \times 10^{-12} \approx 0$
- four suits void: $P(V_1 \cap V_2 \cap V_3 \cap V_4) = 0$

$P(V_1 \cup V_2 \cup V_3 \cup V_4) \approx \binom{4}{3}0.0128 - \binom{4}{2}0.0002 + \binom{4}{3}0$

$\approx 4 \times 0.0128 - 6 \times 0.0002 + 4 \times 0$

$\approx 0.0512 - 0.0012 + 0$

≈ 0.05

Exercise 11.

- toss a fair coin until we observe a head for the first time
- $|\Omega| = 1, 2, 3, 4, ...$
- $p_i = P(i) = (\frac{1}{2})^i$ for i = 1, 2, 3, 4, ...

(a) first head occurs on or before the 3rd toss

- occurs on 1st toss: $\frac{1}{2}$
- occurs on 2nd toss: $(\frac{1}{2})^2 = \frac{1}{4}$
- occurs on 3rd toss: $(\frac{1}{2})^3 = \frac{1}{8}$
- first head occurs on or before the 3rd toss: $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} = \frac{7}{8}$

(b) first head happens after the 5th toss

- occurs on 4th toss: $(\frac{1}{2})^4 = \frac{1}{16}$
- occurs on 5th toss: $(\frac{1}{2})^5 = \frac{1}{32}$
- first head occurs on or before the 5th toss: $\frac{7}{8} + \frac{1}{16} + \frac{1}{32} = \frac{31}{32}$
- first head happens after the 5th toss: $1 - \frac{31}{32} = \frac{1}{32}$

(c) first head happens on a toss which is divisible by 3

- 3, 6, 9, 12, 15, ...
- $\sum_{n=1}^{\infty}(\frac{1}{2})^3 = (\frac{1}{2})^3 + (\frac{1}{2})^6 + (\frac{1}{2})^9 + ...$

  $\frac{\frac{1}{8}}{1-\frac{1}{8}} = \frac{\frac{1}{8}}{\frac{7}{8}} = \frac{1}{8} \times \frac{8}{7} = \frac{1}{7}$

(d) $P(|A|)$ = P(first occurs on or before the 3rd toss) $= \frac{7}{8}$

$P(|B|)$ = P(first head is divisible by 3) $= \frac{1}{7}$

$P(|A \cap B| = \frac{1}{8}$

$P(A|B) = \frac{\frac{1}{8}}{\frac{1}{7}} = \frac{1}{8} \times \frac{7}{1} = \frac{7}{8}$

Exercise 12.

- $P(A \cup B) = 0.7$
- $P(A \cap B^c) = 0.2$
- $P(A^c \cap B) = 0.1$

  $|A \cup B| = |A| + |B| - |A \cup B|$
- $P(A \cup B)$ - A and B
- $P(A \cap B^c)$ - just A
- $P(A^c \cap B)$ - just B
- $P(A) = P(A \cup B) - P(A^c \cap B) = 0.7 - 0.1 = 0.6$
- $P(B) = P(A \cup B) - P(A \cap B^c) = 0.7 - 0.2 = 0.5$
- $P(A \cap B) = |A| + |B| - |A \cup B| = 0.6 + 0.5 - 0.7 = 1.1 - 0.7 = 0.4$

(a) $P(A|A \cup B)| = \frac{|A \cap (A \cup B)|}{|A \cup B|} = \frac{|A|}{|A \cup B|} = \frac{0.6}{0.7} = \frac{6}{7}$

(b) $P(B|A \cup B|) = \frac{|B \cap (A \cup B)|}{|A \cup B|} = \frac{|B|}{|A \cup B|} = \frac{0.5}{0.7} = \frac{5}{7}$

(c) $P(A \cap B|A \cup B) = \frac{|(A \cap B) \cap (A \cup B)|}{|A \cup B|} = \frac{|A \cap B|}{|A \cup B|} = \frac{0.4}{0.7} = \frac{4}{7}$

(d) $P(A|B) = \frac{|A \cap B|}{|B|} = \frac{0.4}{0.5} = \frac{4}{5}$

(e) $P(B|A) = \frac{|A \cap B|}{|A|} = \frac{0.4}{0.6} = \frac{2}{3}$

Exercise 13.

- sequence has 3 of each digit: $\binom{9}{3,3,3} = \frac{9!}{3!3!3!} = 1,680$
- consecutive 1's: $\binom{7}{3,3,1} = 140$

- consecutive 2's: $\binom{7}{3,3,1} = 140$
- consecutive 3's: $\binom{7}{3,3,1} = 140$
- consecutive 1's and 2's: $\binom{5}{3,1,1} = 20$
- consecutive 2's and 3's: $\binom{5}{3,1,1} = 20$
- consecutive 1's and 3's: $\binom{5}{3,1,1} = 20$
- double counting cases: 3! = 6

  111222333

  111333222

  222111333

  222333111

  333111222

  333222111

  => 3 total
- probability that there is at least one digit where all of that digit are consecutive given that a sequence has 3 of each digit $= \frac{3\times140-3\times20+6}{1{,}680} = \frac{366}{1{,}680} = \frac{61}{280} \approx 0.2179$